Department of CSE

Mini Project Report

" Malware Family Classification Using a CNN with Attention on the MaleBin Dataset "

Course Code: ICE 478/CSE475

Section: 2

Submitted to: Dr. Raihan Ul Islam, Associate Professor, Department of Computer Science and Engineering.

Submitted by: Group-1

Student ID	Student Name
2022-3-50-006	Md. Mustafidur Rahman
2022-3-50-008	Amana Akter Arfa
2022-3-50-025	Rakiba Alam

1. Introduction

Malware is one of the major cybersecurity threats in today's digital world. It can damage computer systems, steal sensitive information, and disrupt normal operations. As new malware variants are continuously being developed, traditional signature-based detection methods often fail to identify previously unseen malware. Therefore, researchers are increasingly using machine learning and deep learning techniques for automatic malware detection and classification.

One popular approach is to convert malware binary files into grayscale images and apply image classification techniques. Malware samples from the same family usually produce similar visual patterns, allowing deep learning models such as Convolutional Neural Networks (CNNs) to learn meaningful features directly from the images. In this project, the MaleBin Malware Binary Greyscale Images dataset is used to classify different malware families. Since the project follows Track 3, a CNN with an attention mechanism will be developed in the later stages. Before building the model, it is important to understand the dataset through Exploratory Data Analysis (EDA). This report presents the EDA conducted on the MaleBin dataset and summarizes the key observations obtained from the analysis.

2. Dataset Overview

The MaleBin dataset contains malware binaries that have been converted into grayscale images. Each malware image belongs to a specific malware family, and the images are organized into separate folders according to their labels. The dataset is designed for malware family classification using computer vision techniques.

Unlike conventional malware datasets that require manual feature extraction, this dataset enables deep learning models to automatically learn features from image patterns. However, because malware binaries have different sizes, the generated images also vary in their dimensions. Therefore, preprocessing steps such as image resizing and normalization are necessary before model training.

Dataset Information

- **Dataset Name:** MaleBin Malware Binary Greyscale Images
- **Data Type:** Grayscale Images
- **Image Format:** PNG
- **Classification Type:** Multi-class Malware Family Classification
- **Source:** Kaggle

3. Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was performed to understand the overall characteristics of the dataset before training a deep learning model. The analysis focused on image visualization, class distribution, image dimensions, and pixel intensity patterns. These analyses help identify potential issues such as class imbalance, inconsistent image sizes, and preprocessing requirements.

3.1 Dataset Description

The MaleBin dataset is organized into separate folders, where each folder represents a different malware family. Each folder contains multiple grayscale images generated from malware binary files, and each image belongs to the malware family indicated by its folder name. This folder-based organization makes it convenient to assign class labels during data loading and model training.

Before starting the analysis, the dataset structure was verified to ensure that all images were correctly placed in their respective folders and could be accessed without errors. Understanding this organization is important because it forms the basis for preprocessing, data loading, and supervised learning.

	Property	Value
0	Dataset name	MaleBin (Malware Binary Greyscale Images)
1	Source	Kaggle (tashiee/malebin-malware-binary-greysca...
2	Application area	Malware family classification from binary-to-i...
3	Total images	12464
4	Number of classes	39
5	Image format	PNG/JPG greyscale
6	Balanced?	TBD — see class balance below

Figure 3.1: Dataset Folder Structure

3.2 Sample Malware Images

Random malware images from different classes were visualized to observe their appearance. Although the images are generated from binary files, they exhibit unique texture patterns that differ across malware families. Some images contain dense repetitive structures, while others display irregular patterns.

Visual inspection helps confirm that the images are loaded correctly and provides an initial understanding of the visual differences among malware families.

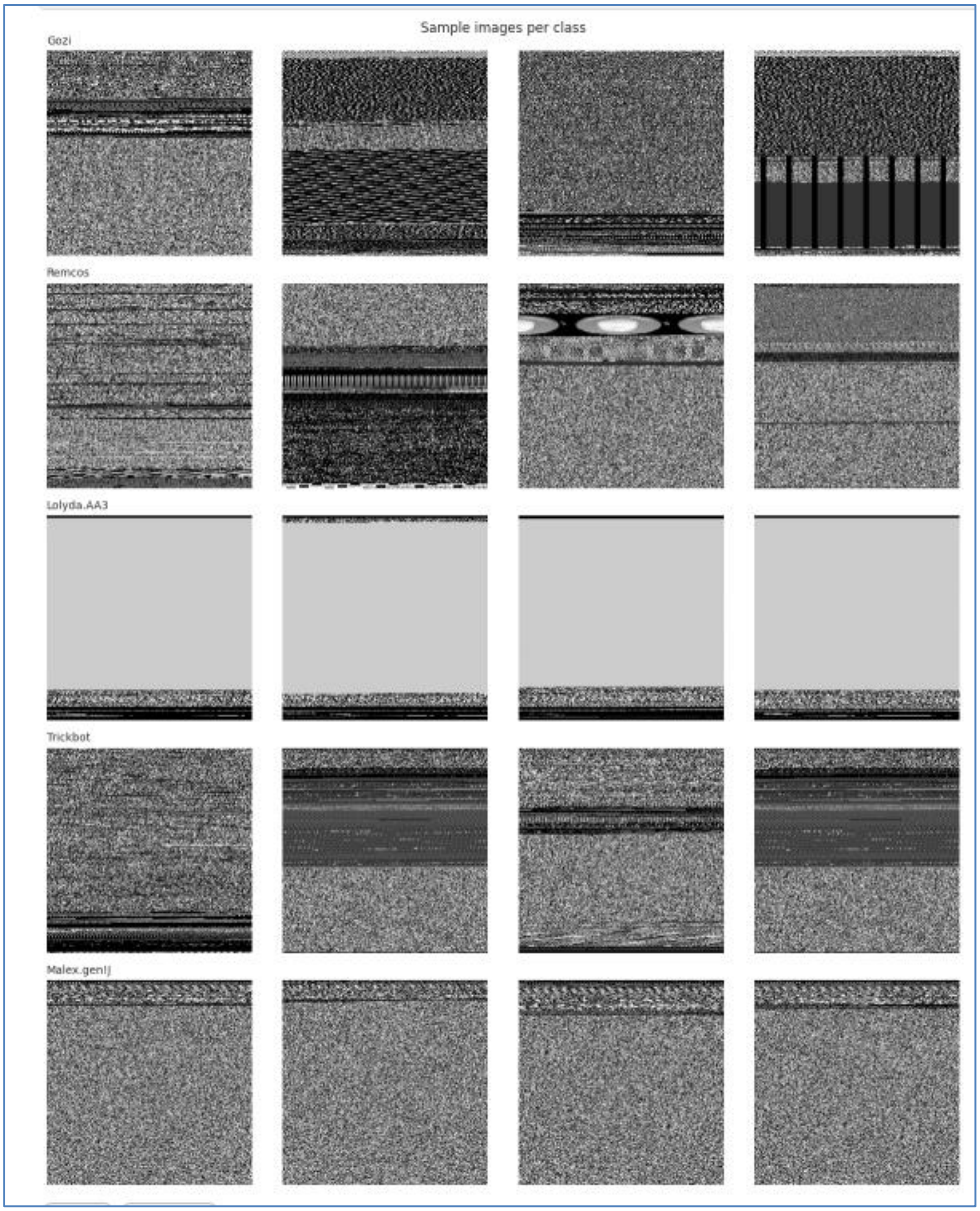

Figure 3.2: Sample Malware Images from Different Malware Families

3.3 Class Distribution

The number of samples in each malware family was analyzed to understand the distribution of the dataset. A bar chart was created to compare the number of images available for each class.

The analysis shows whether the dataset is balanced or imbalanced. A balanced dataset allows the model to learn equally from all classes, while an imbalanced dataset may bias the model toward classes with more samples. If necessary, techniques such as data augmentation or class weighting can be applied during model training.

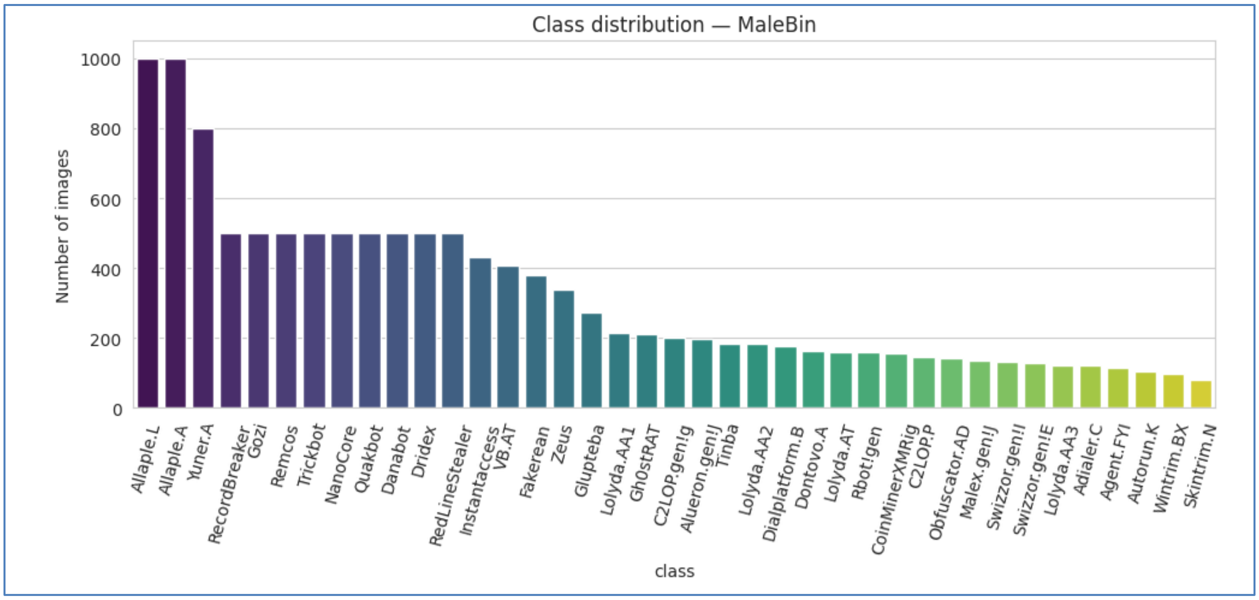

Figure 3.3: Class Distribution of Malware Families

3.4 Image Dimension Analysis

The dimensions of all malware images were analyzed to determine whether they have consistent sizes. Since malware binaries vary in size, the generated images also have different widths and heights.

The analysis indicates that image resizing will be required before training the CNN model. Using a fixed image size ensures that all images can be processed efficiently by the deep learning model.

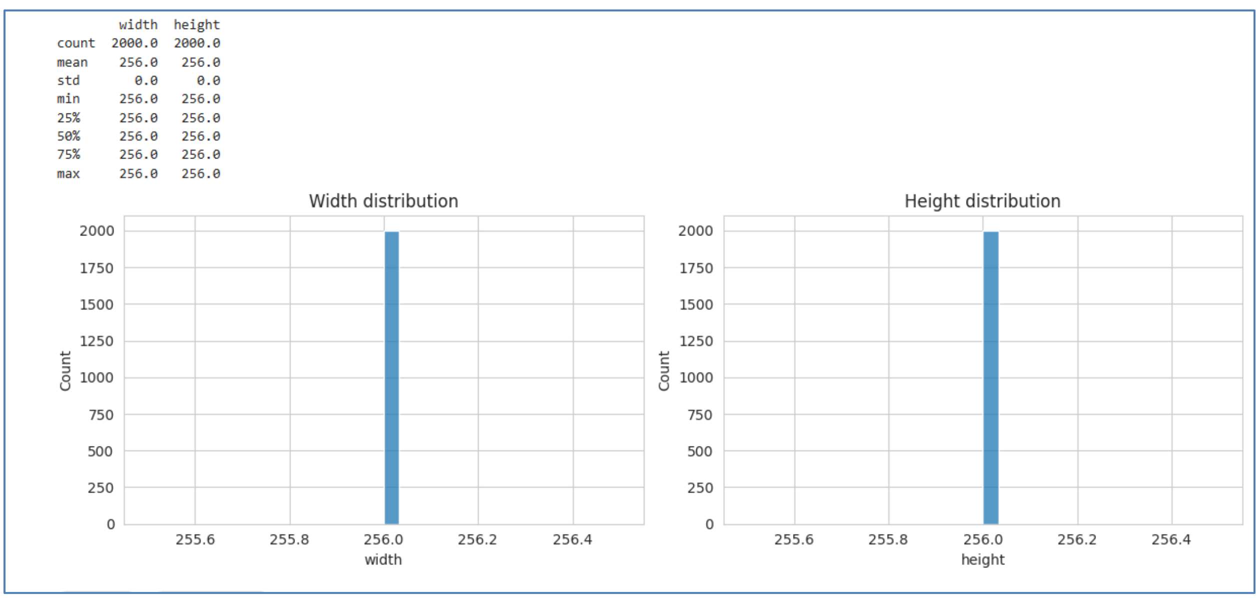

Figure 3.4: Distribution of Image Dimensions

3.5 Pixel Intensity Distribution

Pixel intensity analysis was performed to examine the grayscale value distribution of the malware images. Since the images are represented in grayscale, each pixel has an intensity value ranging from 0 (black) to 255 (white). A histogram was generated to visualize the frequency of these intensity values across the dataset.

The histogram shows that most pixel values are concentrated in the lower and middle intensity ranges, indicating that the malware images contain more dark regions than bright regions. This information is useful for understanding the visual characteristics of the dataset and deciding whether normalization is required before model training.

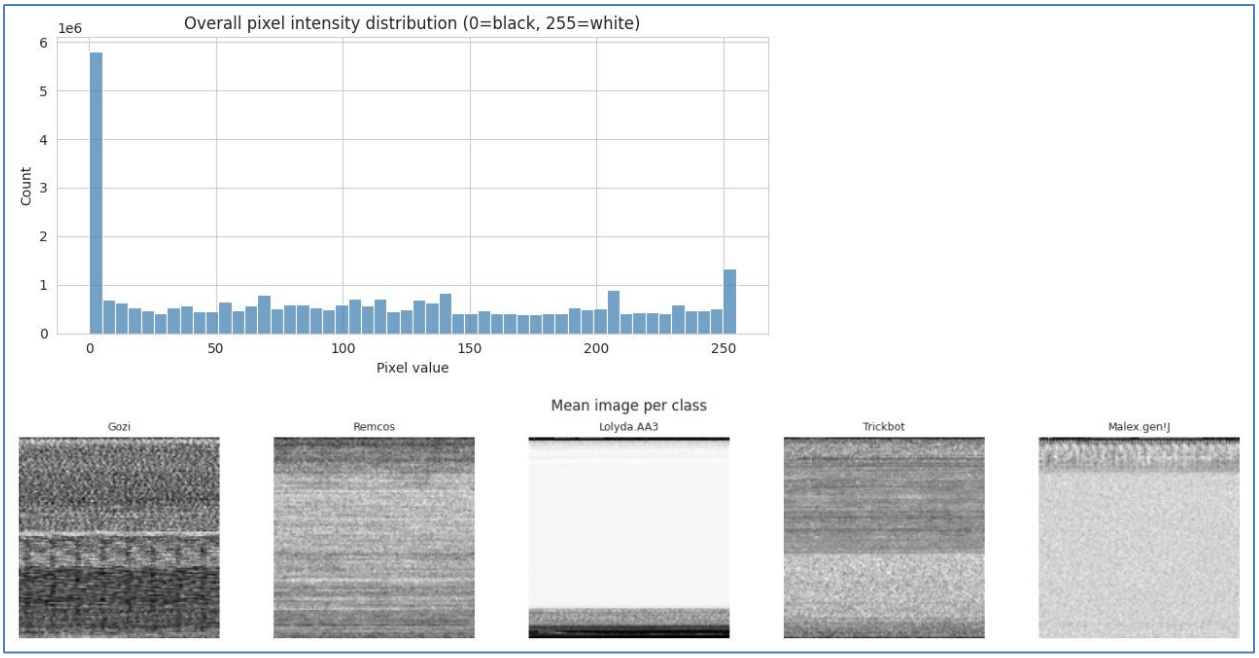

Figure 3.5: Pixel Intensity Distribution of the MaleBin Dataset

3.6 Key Observations

The exploratory data analysis provides several important insights into the MaleBin dataset.

- The dataset contains multiple malware families represented as grayscale images.
- Malware images from the same family show similar visual patterns, while different families exhibit noticeable differences.
- The number of samples varies across malware families, indicating some level of class imbalance.
- Image dimensions are not consistent and require resizing before model training.
- Most pixel intensity values are concentrated in the lower and middle ranges, suggesting that many malware images contain darker regions.

4. Conclusion

This report presented an exploratory analysis of the MaleBin Malware Binary Greyscale Images dataset. The analysis included dataset inspection, visualization of malware images, class distribution, image dimension analysis and pixel intensity distribution.

The results indicate that the dataset is appropriate for malware family classification using deep learning. Although the images have different dimensions, this issue can be addressed through preprocessing techniques such as resizing and normalization. The observed visual patterns also suggest that CNN-based models can effectively learn discriminative features from the malware images.

The insights obtained from this analysis provide a strong foundation for the next stage of the project, where a CNN integrated with an attention mechanism will be developed and evaluated for malware family classification.

References

[1] Tashiee, *MaleBin Malware Binary Greyscale Images Dataset*. Kaggle. Available: https://www.kaggle.com/datasets/tashiee/malebin-malware-binary-greyscale-images

[2] Kaggle Documentation, *Working with Image Datasets*. Available: https://www.kaggle.com/docs